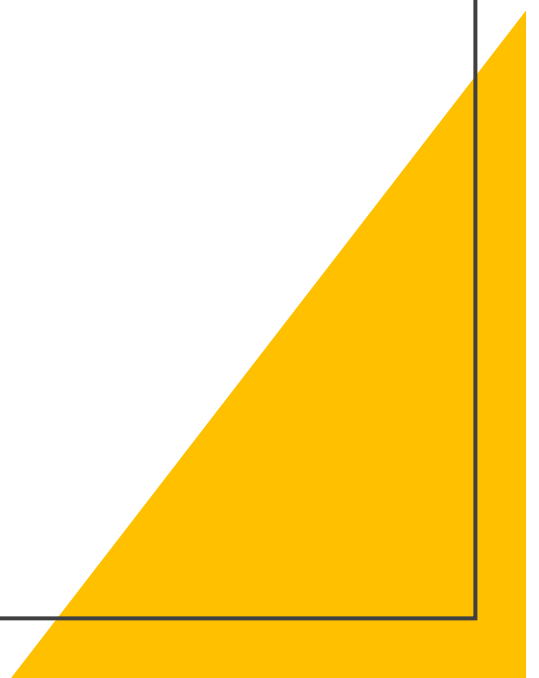


Electron Correlation Methods

Lecture 4

Liviu Ungur



Exact form of the wave function

Exact wave function is obtained when ALL possible Slater Determinants are included in the wave function, while their individual weights are determined variationally:

$$\Psi_{exact} = a_0 \psi_{HF} + \sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r + \sum_{\substack{i,j \\ i < j}}^{N_{occ}} \sum_{\substack{r,s \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs} + \sum_{\substack{i,j,l \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst} + \dots$$

Such wave function is called: exact non-relativistic solution of the Schrodinger equation.

By adding excited Slater determinants into the HF wave function, the electrons start to feel each other – and avoid each other (increasing the inter-electronic distance), accounting for the so-called “electron correlation”.

Such wave functions are only possible for very small molecules, mostly diatomics, with small basis sets. Such wave function is called Full Configuration interaction (FCI).

In practice, it is common to impose a limit on the number of excited electrons compared to the ground state HF Slater Determinant, usually 2.

Hartree Fock SCF Theory (Recap)

- Each electron moves in an average electrostatic field generated by all the remaining electrons and all nuclei (mean field approximation)
- Each molecular (spin-)orbital describes the motion of one single electron (one-electron approximation)
- Multi-electronic wave function Ψ_{HF} is written as an antisymmetrized product of one-electron molecular orbitals (i.e. **a single Slater Determinant**), so that the Pauli exclusion principle is respected
- The Hartree-Fock Energy is obtained as an expectation value of the Hamiltonian:

$$E_{HF} = \int \Psi_{HF}^* \hat{H} \Psi_{HF} d\tau = \langle \Psi_{HF} | \hat{H} | \Psi_{HF} \rangle$$

- Hartree-Fock solution is **higher** in energy compared to the **exact energy**.

$$E_{exact} = \int \Psi_{exact}^* \hat{H} \Psi_{exact} d\tau = \langle \Psi_{exact} | \hat{H} | \Psi_{exact} \rangle$$

- **The difference between exact energy and HF energy is called correlation energy**

$$E_{corr} = E_{exact} - E_{HF}$$

Electron Correlation

Electron correlation is, in effect, defined as inter-electron interactions that Hartree-Fock theory does not properly account for.

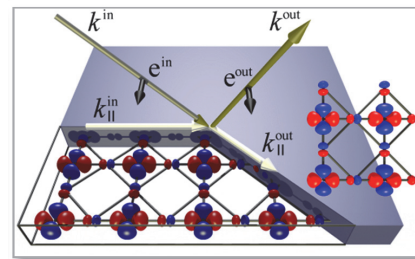
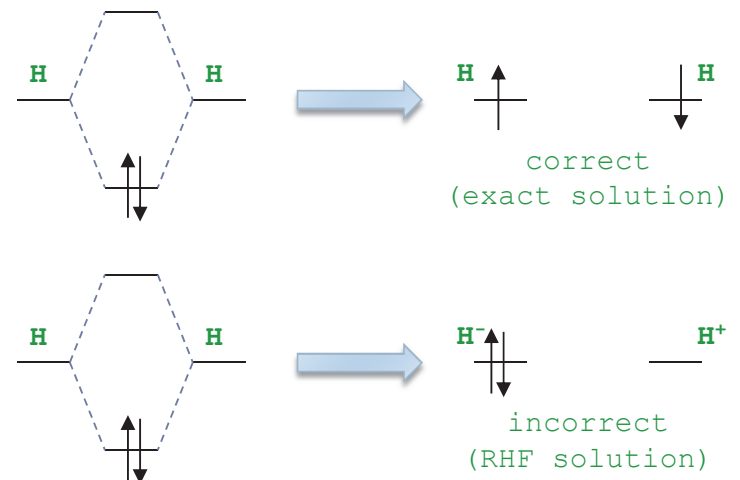
Static correlation refers to a situation where a single Slater determinant (*i.e.* a single population pattern of molecular orbitals) might not be a good approximation to the ground state. This often happens when the ground state is or becomes degenerate.

Example: dissociation of H_2 :

A famous example is the dissociation of the hydrogen molecule, where the restricted Hartree-Fock method produces a pair of ions instead of a pair of radicals.

Dynamic correlation is the correlated *motion* of electrons resulting from the Coulomb repulsion. Being an average field theory, the HF method does not properly account for this effect.

Example: dissociation of H_2



The description of inelastic X-ray scattering cannot be obtained without the inclusion of dynamic correlation effects.

Variational Principle

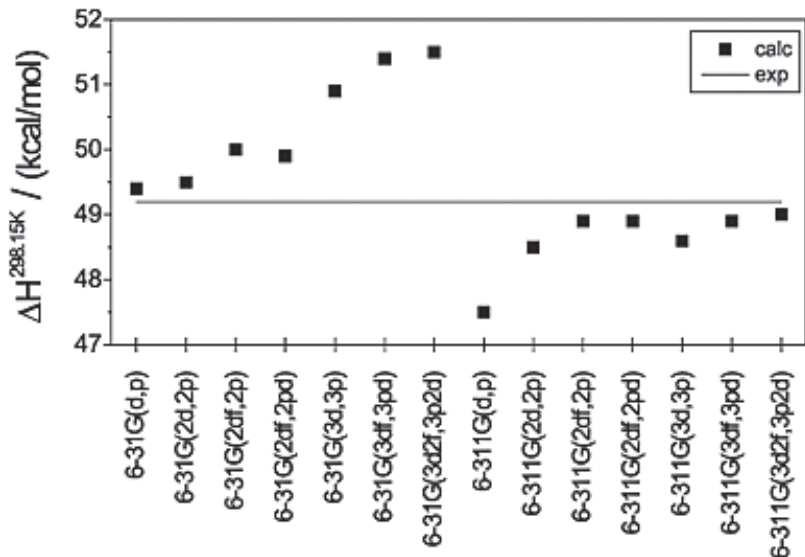
Applied to quantum chemical calculations, the variational principle states that *in a given class of functions, the one with the lowest expectation value for the energy is the best approximation to the ground state of a quantum system.*

In practice, the variational principle guarantees that a bigger basis set would yield a better approximation, though the law of diminishing returns certainly applies.

Basis set	Energy / Hartree
STO-3G	-0.49491
STO-4G	-0.49848
STO-5G	-0.49951
STO-6G	-0.49983

Because of error compensation effects, the variational principle does not generally apply to properties other than energy, and even to energy differences, as the picture on the left demonstrates.

For property calculations, therefore, *a bigger basis set does not necessarily mean a more accurate answer, but certainly means a longer processing time.*



Computational account of electron correlation effects

We have two main avenues that are used in computational chemistry to account for the electron correlation effects:

Multiconfigurational Wave Function Methods:

- Increase the Hartree-Fock wave function by considering additional Slater Determinants (SDs) alongside the original ground state Hartree-Fock Slater Determinant
- Additional Slater Determinants describe various excitation-deexcitation processes accounting for dynamic electron correlation.
- We will further analyse the effects of additional SDs on ensuring the separation between electrons

Density Functional Theory Methods:

- This method describes the exchange and correlation effects in an approximate way. The method still uses the single-Slater Determinant formalism similar to Hartree-Fock, however, with modifies the version of the Fock operator which now accounts for a “softer” potential, which includes the approximate inter-electron correlation term (this term is missing in HF).

Exact form of the wave function

$$\Psi_{exact} = a_0 \psi_{HF} + \underbrace{\sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r}_{\text{single excitations}} + \underbrace{\sum_{\substack{i,j \\ i < j}}^{N_{occ}} \sum_{\substack{r,s \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs}}_{\text{double excitations}} + \underbrace{\sum_{\substack{i,j,l \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst}}_{\text{triple excitations}} + \dots$$

ψ_i^r – a Slater determinant in which the electron from the orbital i is excited towards orbital r

a_i^r – expansion coefficient of this Slater determinant into the full wave function

ψ_{ij}^{rs} – a Slater determinant in which two electrons are excited

...

The *exact wave function* includes ALL possible excited Slater determinants. It can only be computed for very-very small molecules and light atoms, in a small-medium basis set.

It is called sometimes Full Configuration Interaction (FCI) wave function.

Exact form of the wave function

$$\Psi_{exact} = a_0 \psi_{HF} + \sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r + \sum_{\substack{i,j, \\ i < j}}^{N_{occ}} \sum_{\substack{r,s, \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs} + \sum_{\substack{i,j,l \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst} + \dots$$

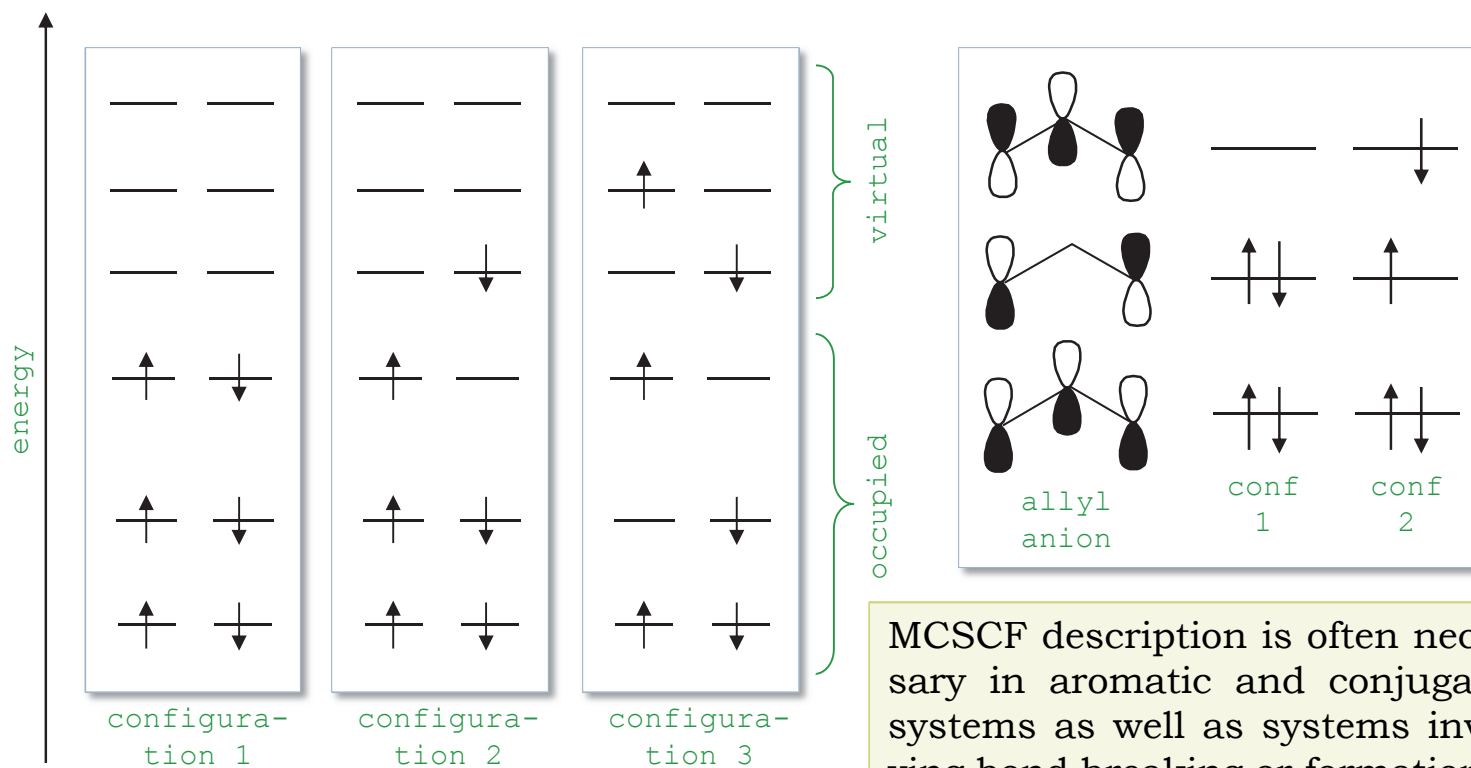
By adding excited Slater determinants into the HF wave function, the electrons start to feel each other – and avoid each other (increasing the inter-electronic distance), accounting for the dynamical correlation.

Two main groups of methods are distinguishable:

- 1) Single-reference methods (MP2, MP3, MP4, CCSD, CCSDT, CISD, CISDT, etc.)
- 2) Multi-reference methods (CASPT2, NEVPT2, etc., MRCI, MRCC, CASSCF, MCSCF, RASSCF, etc.)

Multiconfigurational SCF method

If a single Slater determinant gives a poor description of the system, multiple determinants (differing in the electron arrangement in the orbitals) may be used, with weights adjusted using the *variational principle*.



Configuration Interaction (CI)

We can take MCSCF to the limit and include all excited determinants into the calculation. This is extremely expensive (for combinatorial reasons), and some kind of truncation procedure is often required.

Method	Configurations	Scaling with basis size
CID	all doubly excited determinants	$O(n^5)$
CISD	all singly and doubly excited determinants	$O(n^6)$
CISDT	all singly, doubly and triply excited determinants	$O(n^8)$
full CI	all configurations	$O(n!)$

The limit where full CI is used for a subset of all orbitals is known as CASSCF (complete active space SCF). The choice of the orbitals is not a simple matter.

$$\Psi = a_0 \psi_{HF} + \sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r + \sum_{\substack{i,j \\ i < j}}^{N_{occ}} \sum_{\substack{r,s \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs} + \sum_{\substack{i,j,l \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst} + \dots$$

Due to the astronomical number of configurations, full CI is only possible for very small (diatomic) systems. Fortunately, full CI is rarely necessary, as CISD tends to be sufficiently accurate for most practical purposes.

Configuration Interaction (CI)

$$\Psi = \frac{\text{Diagram 1}}{\text{Diagram 2}} + \sum_{\text{Diagram 3}} \frac{\text{Diagram 4}}{\text{Diagram 5}} + \sum_{\text{Diagram 6}} \frac{\text{Diagram 7}}{\text{Diagram 8}} + \dots$$

The diagram illustrates the Configuration Interaction (CI) expansion of the wavefunction Ψ . It shows a sum of terms, each representing a configuration interaction. The first term is the reference configuration, where all electrons are blue and all holes are white. The subsequent terms are excited configurations, where one or more electrons are promoted to higher energy states, represented by white circles (electrons) and colored circles (holes) in the numerator diagrams. The denominator diagrams represent the reference configuration for each term. The expansion is shown as a sum of these terms, with ellipses indicating further configurations.

Configuration Interaction (CI)

$$\Psi = a_0 \psi_{HF} + \sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r + \sum_{\substack{i,j, \\ i < j}}^{N_{occ}} \sum_{\substack{r,s, \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs} + \sum_{\substack{i,j,l \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t, \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst} + \dots$$

$$= a_0 \psi_{HF} + \sum_S a_S \psi_S + \sum_D a_D \psi_D + \sum_T a_T \psi_T + \dots$$

Optimization of the wave function needs to find $\langle \Psi | \hat{H} | \Psi \rangle \rightarrow \text{minim}$, under constraint $\langle \Psi | \Psi \rangle = 1$.

Proceed to use the **method of Lagrange multipliers**.

- Step 1: Set the *Lagrangian* L :

$$L = \langle \Psi | \hat{H} | \Psi \rangle - \lambda (\langle \Psi | \Psi \rangle - 1) = \sum_{i=0} \sum_{j=0} a_i a_j \langle \psi_i | \hat{H} | \psi_j \rangle - \lambda \left(\sum_{i=0} a_i^2 - 1 \right)$$

- Step 2: Take the derivative of L wrt all parameters, a_i and λ :

$$\frac{\partial L}{\partial a_i} = 2 \sum_{j=0} a_j \langle \psi_i | \hat{H} | \psi_j \rangle - 2\lambda a_i = 2a_i (E_i - \lambda) + 2 \sum_{j \neq i} a_j \langle \psi_i | \hat{H} | \psi_j \rangle = 0$$

Configuration Interaction (CI)

- Step 3: Join all equations for all a_i in a matrix form (*CI Secular Equations*), $\langle \psi_i | \hat{H} | \psi_j \rangle = H_{ij}$:

$$\begin{pmatrix} H_{00} - E & H_{01} & \cdots & H_{0j} & \cdots \\ H_{10} & H_{11} - E & \cdots & H_{1j} & \cdots \\ \vdots & \vdots & \ddots & \vdots & \cdots \\ H_{j0} & H_{j1} & \cdots & H_{jj} - E & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_j \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \vdots \end{pmatrix}$$

- This can be written in a shorthand matrix form:

$$(\mathbf{H} - E\mathbf{I})\mathbf{a} = \mathbf{0}$$

or

$$\mathbf{H}\mathbf{a} = E\mathbf{a}$$

This is the common *eigenvalue problem*. The solution of this equation gives, in principle the exact answer. The only difficulty of applying CI method is the **very fast-growing size of the CI matrix** with the calculated molecule and basis set.

Usually, the truncated CI methods are realized in practice:

CISD= HF + single and double excitations

CISDT= HF + single, double and triple excitations

Slater-Determinant CI and Spin-Adapted CI

Wave functions are usually characterized by a total spin value (S): singlet, triplet, quintet, etc. Significant computational savings can be obtained by combining SDs into spin-adapted Configuration State Functions (CSFs). **States of different spin states do not mix by non-relativistic Hamiltonian:**

$$\langle singlet | \hat{H} | triplet \rangle = 0$$

Number of orbitals $= n$

Number of spin-orbitals $= 2n$

Number of electrons $= N$

Total spin $= S$

$$\text{Number of SD} = C_{2n}^N = \frac{(2n)!}{N! (2n - N)!}$$

$$\text{Number of SD} \geq \text{Number of CSF}$$

$$\text{Number of CSF} = \frac{2S + 1}{n + 1} \times C_{n+1}^{\frac{1}{2}N - S} \times C_{n+1}^{\frac{1}{2}N + S + 1} =$$

$$= \frac{2S + 1}{n + 1} \times \frac{(n + 1)!}{\left(\frac{1}{2}N - S\right)! \left(n + 1 - \frac{1}{2}N + S\right)!} \times \frac{(n + 1)!}{\left(\frac{1}{2}N + S + 1\right)! \left(n - \frac{1}{2}N - S\right)!}$$

Computational difficulty of CI

Excitation level n	Number of n th excited CSFs	Total number of CSFs
1	71	71
2	2 485	2 556
3	40 040	42 596
4	348 530	391 126
5	1 723 540	2 114 666
6	5 033 210	7 147 876
7	8 688 680	15 836 556
8	8 653 645	24 490 201
9	4 554 550	29 044 751
10	1 002 001	30 046 752

Weight of excitation level for Ne in [5s4p3d] basis

Excitation level	Weight
0	9.6×10^{-1}
1	9.8×10^{-4}
2	3.4×10^{-2}
3	3.7×10^{-4}
4	4.5×10^{-4}
5	1.9×10^{-5}
6	1.7×10^{-6}
7	1.4×10^{-7}
8	1.1×10^{-9}



How CI method allows for electrons to be far apart from each other?

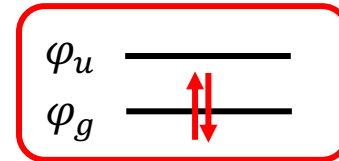
Consider H_2 molecule in **minimal basis set** (use only 1 function localized on each atom):

$$\varphi_g = (\chi_A + \chi_B)$$

$$\varphi_u = (\chi_A - \chi_B)$$

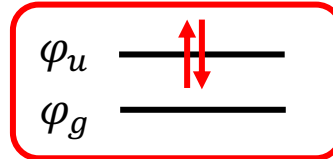
Ground state SD:

$$\Psi_0 = \begin{vmatrix} \varphi_g(1) & \overline{\varphi_g}(1) \\ \varphi_g(2) & \overline{\varphi_g}(2) \end{vmatrix}$$

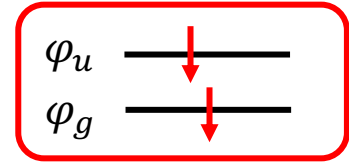
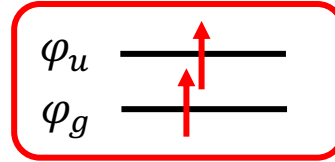
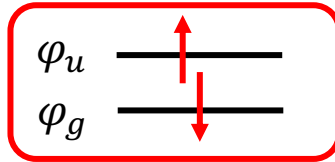
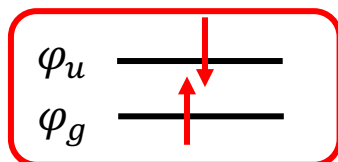


Full CI matrix in this basis contains Ψ_0 , four singly-excited SDs and one doubly-excited SD:

$$\Psi_1 = \begin{vmatrix} \varphi_u(1) & \overline{\varphi_u}(1) \\ \varphi_u(2) & \overline{\varphi_u}(2) \end{vmatrix} \text{ (doubly excited)}$$



$$\Psi_2 = \begin{vmatrix} \varphi_g(1) & \overline{\varphi_u}(1) \\ \varphi_g(2) & \overline{\varphi_u}(2) \end{vmatrix}; \quad \Psi_3 = \begin{vmatrix} \overline{\varphi_g}(1) & \varphi_u(1) \\ \overline{\varphi_g}(2) & \varphi_u(2) \end{vmatrix}; \quad \Psi_4 = \begin{vmatrix} \varphi_g(1) & \varphi_u(1) \\ \varphi_g(2) & \varphi_u(2) \end{vmatrix}; \quad \Psi_5 = \begin{vmatrix} \overline{\varphi_g}(1) & \overline{\varphi_u}(1) \\ \overline{\varphi_g}(2) & \overline{\varphi_u}(2) \end{vmatrix}$$



CI matrix for H₂ in minimal basis

By **spin adaptation of the wave functions**, we can separate *spin triplet* and *spin singlet* states:

$$S = 0: \quad \Psi_0, \quad \Psi_1, \quad (\Psi_2 - \Psi_3)$$

$$S = 1: \quad \Psi_4, \quad \Psi_5, \quad (\Psi_2 + \Psi_3)$$

The CI matrix is:

$$\begin{array}{c}
 \Psi_0 \\
 \Psi_1 \\
 (\Psi_2 - \Psi_3) \\
 \hline
 \Psi_4 \\
 (\Psi_2 + \Psi_3) \\
 \Psi_5
 \end{array}
 \left|
 \begin{array}{ccc|ccc}
 E_0 & H_{01} & 0 & 0 & 0 & 0 \\
 H_{10} & E_1 & 0 & 0 & 0 & 0 \\
 0 & 0 & E_2 & 0 & 0 & 0 \\
 \hline
 0 & 0 & 0 & E_3 & 0 & 0 \\
 0 & 0 & 0 & 0 & E_4 & 0 \\
 0 & 0 & 0 & 0 & 0 & E_5
 \end{array}
 \right)
 \begin{array}{c}
 \leftarrow S=0 \\
 \\
 \leftarrow S=1 \\
 \\
 \begin{array}{cc}
 \boxed{S=0} & \boxed{S=1}
 \end{array}
 \end{array}$$

Spin singlet states do NOT mix with spin triplet states. Also, singly-excited singlet SD ($\Psi_2 - \Psi_3$) does not mix with Ψ_0 because of Brillouin theorem (all singly-excited configurations do not mix with Hartree-Fock ground state).

CI matrix for H₂ in minimal basis

As result, the diagonalization of the CI matrix leads to a mixing between Ψ_0 (ground state at HF) and doubly-excited Ψ_1 .

$$\begin{aligned}\Psi_0 &= \begin{vmatrix} \varphi_g(1) & \overline{\varphi_g}(1) \\ \varphi_g(2) & \overline{\varphi_g}(2) \end{vmatrix} = \varphi_g(1)\overline{\varphi_g}(2) - \overline{\varphi_g}(1)\varphi_g(2) = \varphi_g(1)\varphi_g(2)(\alpha\beta - \beta\alpha) = \\ &= (\chi_A(1) + \chi_B(1))(\chi_A(2) + \chi_B(2)) = \underbrace{\chi_A(1)\chi_A(2) + \chi_B(1)\chi_B(2)}_{\text{ionic terms}} + \underbrace{\chi_A(1)\chi_B(2) + \chi_B(1)\chi_A(2)}_{\text{covalent terms}}\end{aligned}$$

$$\begin{aligned}\Psi_1 &= \begin{vmatrix} \varphi_u(1) & \overline{\varphi_u}(1) \\ \varphi_u(2) & \overline{\varphi_u}(2) \end{vmatrix} = \varphi_u(1)\overline{\varphi_u}(2) - \overline{\varphi_u}(1)\varphi_u(2) = \varphi_u(1)\varphi_u(2)(\alpha\beta - \beta\alpha) \\ &= (\chi_A(1) - \chi_B(1))(\chi_A(2) - \chi_B(2)) = \chi_A(1)\chi_A(2) + \chi_B(1)\chi_B(2) - \chi_A(1)\chi_B(2) - \chi_B(1)\chi_A(2)\end{aligned}$$

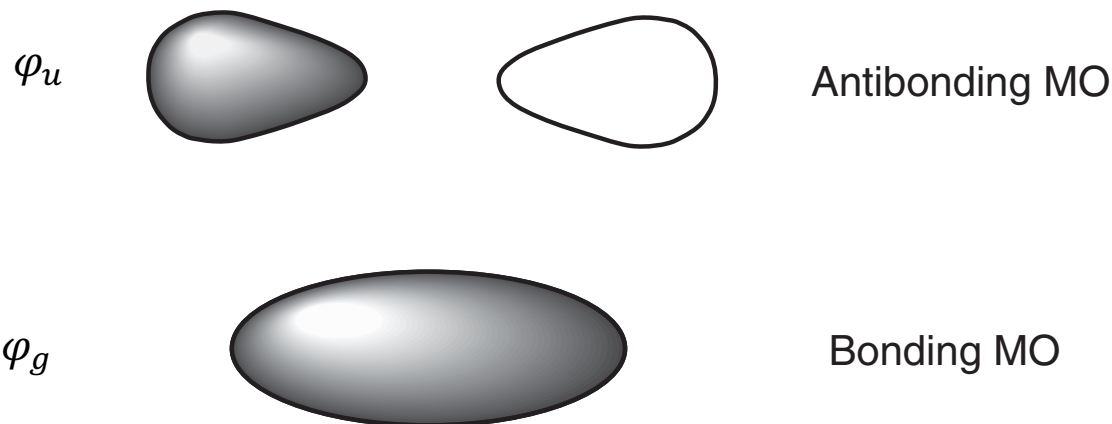
The Full CI is a linear combination between Ψ_0 and Ψ_1 :

$$\Psi_{FCI} = a_0\Psi_0 + a_1\Psi_1, \text{ with } a_0^2 + a_1^2 = 1$$

$$\Psi_{FCI} = (a_0 + a_1)(\chi_A(1)\chi_A(2) + \chi_B(1)\chi_B(2)) + (a_0 - a_1)(\chi_A(1)\chi_B(2) + \chi_B(1)\chi_A(2))$$

The HF wave function Ψ_0 has 50% ionic and 50% covalent contributions. By admixing the Ψ_1 into Ψ_0 , the **covalent** contribution usually increases.

Basis Set approximation

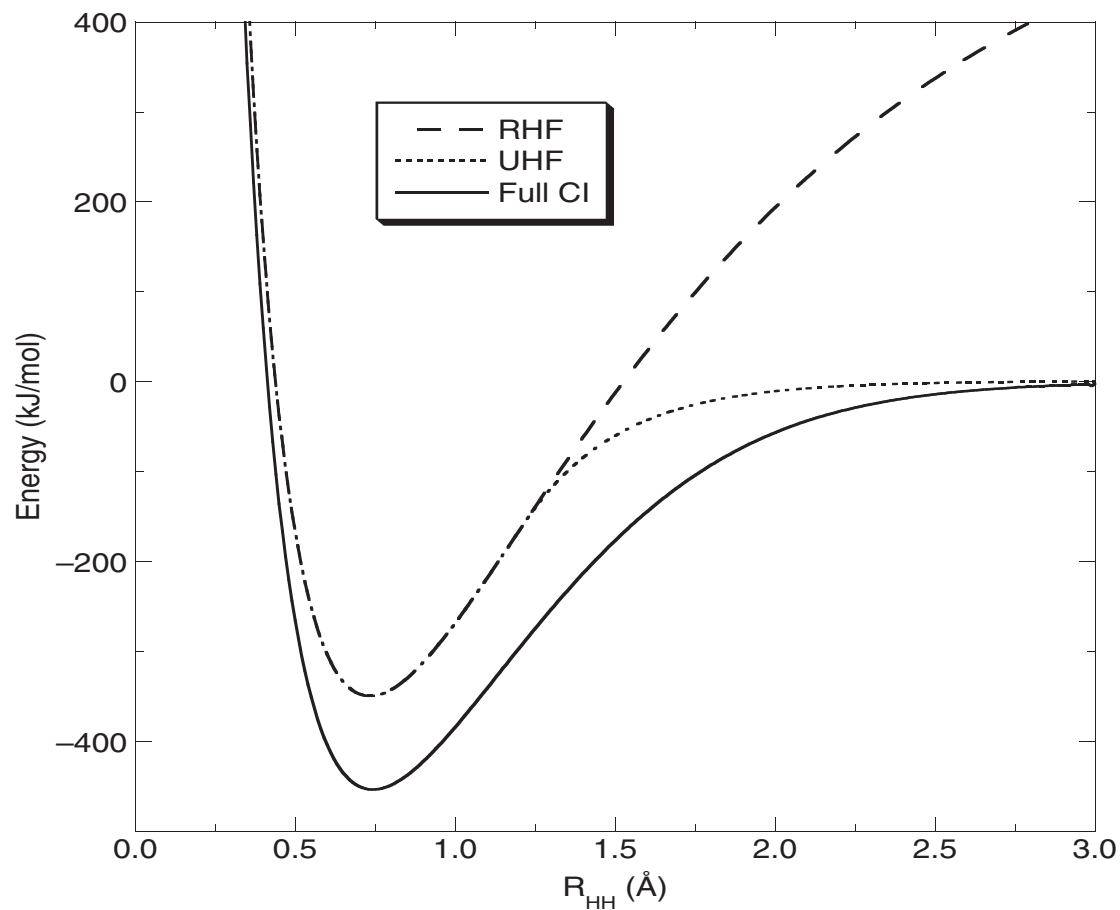


$$\Psi_{FCI} = (a_0 + a_1)(\chi_A(1)\chi_A(2) + \chi_B(1)\chi_B(2)) + (a_0 - a_1)(\chi_A(1)\chi_B(2) + \chi_B(1)\chi_A(2))$$

The HF wave function Ψ_0 has 50% ionic and 50% covalent contributions. By admixing the Ψ_1 into Ψ_0 , the **covalent** contribution usually increases. This allows for a correlation of the motion of the two electrons (left-right correlation).

The % of admixture of excited configurations depends strongly on the H-H distance. At infinite large distance $a_1 = -a_0 \rightarrow$ ionic term becomes zero!

Comparison between RHF, UHF and FCI for H₂



Size Inconsistency of truncated CI methods

Let's assume we need to compute the energy of two water molecules, separated by e.g. 100 Angstrom. We can perform 2 calculations:

- 1) Compute *both water molecules together*, in the same calculation (assign basis sets to all 2O and 4 H) → get total energy E_{full} .
- 2) Compute *one single water* molecule and then, multiply the result by 2. Get the total energy E_{sum} .

We would expect that $E_{full} = E_{sum}$. However, this equality is respected only by the Full CI method (when all excitations are included in both calculations).

When using any of the truncated CI methods (like CISD), we will find that $E_{sum} < E_{full}$. The reason for this behavior arises from the fact that the E_{full} included only single- and double excitations for the entire basis set of both molecules, while E_{sum} includes single and double excitations for each fragment only. When we multiply by 2 the energy of the one water molecule, we essentially take the product of two CISD wave functions of each water molecule. Such a product of two CISD wave functions will necessarily include single, double, triple and quadruple excited SDs, more than the CISD wave function obtained for the case (1).

Coupled Clusters (CC)

$$\Psi = \frac{\text{Diagram 1}}{\text{Diagram 2}} + \sum_{\text{Diagram 3}} \frac{\text{Diagram 4}}{\text{Diagram 5}} + \sum_{\text{Diagram 6}} \frac{\text{Diagram 7}}{\text{Diagram 8}} + \dots$$

Coupled-cluster theory uses a sum over *all* excitations of a *specified type*. That is, whereas each determinant is parameterized in CI theory, it is the excitation process that is parameterized in the CC theory. CC includes higher excitations, but their coefficients are products of lower order excitation coefficients.

CC theory is size-consistent, that is, for two systems: $E_{CC}(1 + 2) = E_{CC}(1) + E_{CC}(2)$

Coupled Clusters (CC)

Coupled Clusters method (like CI) aims at adding excited Slater Determinants into the ground state HF wave function. The difference between CC and CI lies in the selection of which excited SD are included in the wave function.

$$T = T_1 + T_2 + T_3 + \cdots + T_{N_{elec}}$$

T_1 —operator allowing to generate singly-excited configurations

T_2 —operator allowing to generate doubly-excited configurations

The CI wave function can be written:

$$\Psi_{CI} = (1 + T)\Psi_0 = (1 + T_1 + T_2 + T_3 + \cdots + T_{N_{elec}})\Psi_0$$

The corresponding Coupled Clusters wave function is defined as:

$$\begin{aligned}\Psi_{CC} &= e^T \Psi_0 \Rightarrow (\text{Taylor expansion of } e^T) \\ &= \left[1 + \textcolor{red}{T}_1 + \left(T_2 + \frac{1}{2} T_1^2 \right) + \left(T_3 + T_2 T_1 + \frac{1}{6} T_1^3 \right) + \cdots \right] \Psi_0\end{aligned}$$

By comparison with the CI wave function, the CC wave function at each excitation level contains additional terms arising from products of excitations. Schrodinger equation becomes:

$$\mathbf{H}\Psi_{CC} = E_{CC}\Psi_{CC} \rightarrow \mathbf{H}e^T\Psi_0 = E_{CC}e^T\Psi_0$$

Coupled Clusters (CC)

$$\mathbf{H}\Psi_{CC} = E_{CC}\Psi_{CC} \rightarrow \mathbf{H}e^T\Psi_0 = E_{CC}e^T\Psi_0$$

By multiplying Ψ_0^* and integrate over all space, we get:

$$\langle\Psi_0|\mathbf{H}e^T|\Psi_0\rangle = E_{CC}\langle\Psi_0|e^T\Psi_0\rangle = E_{CC}\langle\Psi_0|(1 + T_1 + T_2 + T_3 + \cdots + T_{Nelec})\Psi_0\rangle = E_{CC}$$

Thus, in order to get E_{CC} , we need to evaluate one integral (here we assume the truncation the exponent at second order):

$$\begin{aligned} E_{CC} &= \langle\Psi_0|\mathbf{H}e^T|\Psi_0\rangle = \left\langle\Psi_0\left|\mathbf{H}\left(1 + T_1 + T_2 + \frac{1}{2}T_1^2\right)\right|\Psi_0\right\rangle = \\ &= \langle\Psi_0|\mathbf{H}|\Psi_0\rangle + \langle\Psi_0|\mathbf{H}|T_1\Psi_0\rangle + \langle\Psi_0|\mathbf{H}|T_2\Psi_0\rangle + \frac{1}{2}\langle\Psi_0|\mathbf{H}|T_1^2\Psi_0\rangle = \\ &= E_0 + \sum_i^{occ} \sum_a^{virt} t_i^a \langle\Psi_0|\mathbf{H}|\Psi_i^a\rangle + \sum_{i<j}^{occ} \sum_{a<b}^{virt} (t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a) \langle\Psi_0|\mathbf{H}|\Psi_{ij}^{ab}\rangle \\ &= E_0 + \sum_{i<j}^{occ} \sum_{a<b}^{virt} (t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a) \langle\Psi_0|\mathbf{H}|\Psi_{ij}^{ab}\rangle \end{aligned}$$

from Brillouin Theorem

$$\sum_i^{occ} \sum_a^{virt} t_i^a \langle\Psi_0|\mathbf{H}|\Psi_i^a\rangle = 0$$

Coupled Clusters (CC)

$$\begin{aligned}
 E_{CC} &= E_0 + \sum_{i < j}^{occ} \sum_{a < b}^{virt} (t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a) \langle \Psi_0 | \mathbf{H} | \Psi_{ij}^{ab} \rangle \\
 &= E_0 + \sum_{i < j}^{occ} \sum_{a < b}^{virt} (t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a) (\langle \varphi_i \varphi_j | \varphi_a \varphi_b \rangle - \langle \varphi_i \varphi_j | \varphi_b \varphi_a \rangle)
 \end{aligned}$$

To evaluate the above expression, we need to know amplitudes: $t_{ij}^{ab}, t_i^a t_j^b, t_i^b t_j^a$ and the transformed integrals to MO basis: $\langle \varphi_i \varphi_j | \varphi_a \varphi_b \rangle, \langle \varphi_i \varphi_j | \varphi_b \varphi_a \rangle$.

Amplitudes $t_{ij}^{ab}, t_i^a t_j^b, t_i^b t_j^a$ are obtained from solving such equations:

$$\begin{aligned}
 \langle \Psi_i^a | \mathbf{H} e^T | \Psi_0 \rangle &= E_{CC} \langle \Psi_i^a | e^T | \Psi_0 \rangle \\
 \langle \Psi_{ij}^{ab} | \mathbf{H} e^T | \Psi_0 \rangle &= E_{CC} \langle \Psi_{ij}^{ab} | e^T | \Psi_0 \rangle
 \end{aligned}$$

In practice, the CC method is truncated at a certain rank:

CCSD = Coupled Clusters singles and doubles

CCSDT = Coupled Clusters singles, doubles and triples → golden standard in QC is CCSDTQ

Coupled Clusters is size-consistent method, even at the truncated level.

Coupled Clusters Theory

Method	RMS bond length error, (Angstrom from XRD)
HF/6-31G(d,p)	0.021
HF/6-311G(d,p)	0.022
MP2/6-31G(d,p)	0.014
MP2/6-311G(d,p)	0.014
QCISD/6-311G(d,p)	0.013
CCSD(T)/6-311G(d,p)	0.013

Method	RMS atomization energy error, (kcal/mol from experiment)
HF/6-31G(d)	85.9
HF/6-311G(2df,p)	82.0
MP2/6-31G(d)	22.4
MP2/6-311G(d,p)	23.7
QCISD/6-311G(d)	28.8
CCSD(T)/6-311G(2df,p)	11.5

Method	$E_{CC}-E_{FCI}$ for water in cc-pVDZ, Hartree
RHF	0.217822
CCSD	0.003744
CCSDT	0.000493
CCSDTQ	0.000019
CCSDTQP	0.000003

Convergence with excitation level may be much slower in multi-reference systems (such as twisted double bonds, stretched single bonds, *etc.*) – just as with CI, if a single determinant is not a good first approximation to the ground state, CASSCF / CASPT methods should be used instead.

Perturbation theory

- Perturbation theory is one of the most important and useful methods in physics in general, and quantum mechanics in particular, allowing to treat complicated Hamiltonians of complex systems. For simplicity, we will outline the basic ideas of this method.
- Assume a simpler Hamiltonian $\hat{H}_0$ whose wave function and energy at particular state is known. We have:

$$\hat{H}_0 \left| \psi_n^{(0)} \right\rangle = E_n^{(0)} \left| \psi_n^{(0)} \right\rangle$$

- Suppose that a new Hamiltonian can be written as:

$$\hat{H} = \hat{H}_0 + \lambda \hat{H}_1$$

- As example, $\hat{H}_0 = \hat{K} + \hat{V}_0$. We assume that the interaction force $\hat{V}_0$ is much stronger than the perturbation force $\lambda \hat{H}_1$.

Perturbation theory

- $\hat{H}_1$ can be e.g. an external electric or magnetic field. The action of $\hat{H}_1$ on the original wave function $|\psi_n^{(0)}\rangle$ will change the state and its energy.
- The new (perturbed) wave function of the n th state can be written:

$$|\psi_n\rangle = \sum_{k=0}^{\infty} \lambda^k |\psi_n^{(k)}\rangle, \quad n = 1, 2, 3, \dots$$

And the corresponding energy is:

$$E_n = \sum_{k=0}^{\infty} \lambda^k E_n^{(k)}, \quad n = 1, 2, 3, \dots$$

where $|\psi_n^{(k)}\rangle$ and $E_n^{(k)}$ are the k^{th} order corrections to the wave function and energy

The full Hamiltonian becomes:

$$\hat{H}\psi_n = (\hat{H}_0 + \lambda\hat{H}_1)\psi_n = E_n \psi_n$$

Substitute the expressions for the $|\psi_n\rangle$ and E_n here:

$$(\hat{H}_0 + \lambda\hat{H}_1) \sum_{k=0}^{\infty} \lambda^k |\psi_n^{(k)}\rangle = \left(\sum_{k=0}^{\infty} \lambda^k E_n^{(k)} \right) \left(\sum_{k=0}^{\infty} \lambda^k |\psi_n^{(k)}\rangle \right)$$

Write down explicitly terms with powers of λ smaller than 2, also combining terms which have the same power of λ :

$$\begin{aligned} & \hat{H}_0 |\psi_n^{(0)}\rangle + \lambda (\hat{H}_0 |\psi_n^{(1)}\rangle + \hat{H}_1 |\psi_n^{(0)}\rangle) + \lambda^2 (\hat{H}_0 |\psi_n^{(2)}\rangle + \hat{H}_1 |\psi_n^{(1)}\rangle) + \dots = \\ & = E_n^{(0)} |\psi_n^{(0)}\rangle + \lambda (E_n^{(0)} |\psi_n^{(1)}\rangle + E_n^{(1)} |\psi_n^{(0)}\rangle) + \lambda^2 (E_n^{(0)} |\psi_n^{(2)}\rangle + E_n^{(1)} |\psi_n^{(1)}\rangle + E_n^{(2)} |\psi_n^{(0)}\rangle) + \dots \end{aligned}$$

Equalizing the terms with similar powers of λ :

$$\hat{H}_0 \left| \psi_n^{(0)} \right\rangle = E_n^{(0)} \left| \psi_n^{(0)} \right\rangle$$

$$\hat{H}_0 \left| \psi_n^{(1)} \right\rangle + \hat{H}_1 \left| \psi_n^{(0)} \right\rangle = E_n^{(0)} \left| \psi_n^{(1)} \right\rangle + E_n^{(1)} \left| \psi_n^{(0)} \right\rangle$$

$$\hat{H}_0 \left| \psi_n^{(2)} \right\rangle + \hat{H}_1 \left| \psi_n^{(1)} \right\rangle = E_n^{(0)} \left| \psi_n^{(2)} \right\rangle + E_n^{(1)} \left| \psi_n^{(1)} \right\rangle + E_n^{(2)} \left| \psi_n^{(0)} \right\rangle$$

...

Multiply the second equation by $\left\langle \psi_m^{(0)} \right|$:

$$\left\langle \psi_m^{(0)} \right| \hat{H}_0 \left| \psi_n^{(1)} \right\rangle + \left\langle \psi_m^{(0)} \right| \hat{H}_1 \left| \psi_n^{(0)} \right\rangle = \left\langle \psi_m^{(0)} \right| E_n^{(0)} \left| \psi_n^{(1)} \right\rangle + \left\langle \psi_m^{(0)} \right| E_n^{(1)} \left| \psi_n^{(0)} \right\rangle$$

We also notice:

$$\left\langle \psi_m^{(0)} \right| \hat{H}_0 \left| \psi_n^{(1)} \right\rangle = \left\langle \psi_n^{(1)} \right| \hat{H}_0 \left| \psi_m^{(0)} \right\rangle^* = \left\langle \psi_n^{(1)} \right| \hat{H}_0 \psi_m^{(0)} \right\rangle^* = E_m^{(0)*} \left\langle \psi_n^{(1)} \right| \psi_m^{(0)} \right\rangle^* = E_m^{(0)} \left\langle \psi_m^{(0)} \right| \psi_n^{(1)} \right\rangle$$

$$E_m^{(0)} \langle \psi_m^{(0)} | \psi_n^{(1)} \rangle - E_n^{(0)} \langle \psi_m^{(0)} | \psi_n^{(1)} \rangle = E_n^{(1)} \delta_{mn} - \langle \psi_m^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle$$

If $m = n$ the left hand side is 0:

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle$$

$$E_n \approx E_n^{(0)} + E_n^{(1)} = E_n^{(0)} + \langle \psi_n^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle$$

If $m \neq n$ the above equation becomes:

$$(E_m^{(0)} - E_n^{(0)}) \langle \psi_m^{(0)} | \psi_n^{(1)} \rangle = - \langle \psi_m^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle$$

We can expand the $\psi_n^{(1)}$ in a complete set of unperturbed functions $\psi_n^{(0)}$:

$$\psi_n^{(1)} = \sum_m a_m \psi_m^{(0)}; \quad \Rightarrow \quad a_m = \langle \psi_m^{(0)} | \psi_n^{(1)} \rangle$$

Substitute this form $\psi_n^{(1)}$ in the above equation:

$$\left(E_m^{(0)} - E_n^{(0)}\right) a_m = -\left\langle \psi_m^{(0)} \left| \hat{H}_1 \right| \psi_n^{(0)} \right\rangle ; \quad a_m = \frac{\left\langle \psi_m^{(0)} \left| \hat{H}_1 \right| \psi_n^{(0)} \right\rangle}{\left(E_n^{(0)} - E_m^{(0)}\right)} ;$$

$$\psi_n^{(1)} = \sum_m a_m \psi_m^{(0)} = \sum_m \frac{\left\langle \psi_m^{(0)} \left| \hat{H}_1 \right| \psi_n^{(0)} \right\rangle}{\left(E_n^{(0)} - E_m^{(0)}\right)} \psi_m^{(0)}$$

Full wave function:

$$\psi_n = \psi_n^{(0)} + \sum_m \frac{\left\langle \psi_m^{(0)} \left| \hat{H}_1 \right| \psi_n^{(0)} \right\rangle}{\left(E_n^{(0)} - E_m^{(0)}\right)} \psi_m^{(0)}$$

Select the equation corresponding to λ^2

$$\hat{H}_0 \left| \psi_n^{(2)} \right\rangle + \hat{H}_1 \left| \psi_n^{(1)} \right\rangle = E_n^{(0)} \left| \psi_n^{(2)} \right\rangle + E_n^{(1)} \left| \psi_n^{(1)} \right\rangle + E_n^{(2)} \left| \psi_n^{(0)} \right\rangle$$

Multiply it with $\left\langle \psi_m^{(0)} \right|$:

$$\left\langle \psi_m^{(0)} \right| \hat{H}_0 \left| \psi_n^{(2)} \right\rangle + \left\langle \psi_m^{(0)} \right| \hat{H}_1 \left| \psi_n^{(1)} \right\rangle = \left\langle \psi_m^{(0)} \right| E_n^{(0)} \left| \psi_n^{(2)} \right\rangle + \left\langle \psi_m^{(0)} \right| E_n^{(1)} \left| \psi_n^{(1)} \right\rangle + \left\langle \psi_m^{(0)} \right| E_n^{(2)} \left| \psi_n^{(0)} \right\rangle$$

As before, we find that: $\left\langle \psi_m^{(0)} \right| \hat{H}_0 \left| \psi_n^{(2)} \right\rangle = E_m^{(0)} \left\langle \psi_m^{(0)} \right| \psi_n^{(2)} \right\rangle$

$$\left(E_m^{(0)} - E_n^{(0)} \right) \left\langle \psi_m^{(0)} \right| \psi_n^{(2)} \right\rangle = E_n^{(2)} \delta_{mn} + E_n^{(1)} \left\langle \psi_m^{(0)} \right| \psi_n^{(1)} \right\rangle - \left\langle \psi_m^{(0)} \right| \hat{H}_1 \left| \psi_n^{(1)} \right\rangle$$

If $m = n$ the left-hand side is 0, and since $\left\langle \psi_n^{(0)} \right| \psi_n^{(1)} \right\rangle = 0$:

$$E_n^{(2)} = \left\langle \psi_n^{(0)} \right| \hat{H}_1 \left| \psi_n^{(1)} \right\rangle$$

$$E_n^{(2)} = \left\langle \psi_n^{(0)} \left| \hat{H}_1 \right| \psi_n^{(1)} \right\rangle$$

Substitute the $\psi_n^{(1)} = \sum_m \frac{\langle \psi_m^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle}{(E_n^{(0)} - E_m^{(0)})} \psi_m^{(0)}$ in the above energy expression to find:

$$E_n^{(2)} = \left\langle \psi_n^{(0)} \left| \hat{H}_1 \right| \psi_n^{(1)} \right\rangle = \sum_m \frac{\langle \psi_m^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle}{(E_n^{(0)} - E_m^{(0)})} \langle \psi_n^{(0)} | \hat{H}_1 | \psi_m^{(0)} \rangle$$

$$E_n^{(2)} = \sum_m \frac{\left| \langle \psi_m^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle \right|^2}{(E_n^{(0)} - E_m^{(0)})}$$

$$E_n \approx E_n^{(0)} + E_n^{(1)} + E_n^{(2)} = E_n^{(0)} + \langle \psi_n^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle + \sum_m \frac{\left| \langle \psi_m^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle \right|^2}{(E_n^{(0)} - E_m^{(0)})}$$

Moller-Plesset Perturbation theory

It is the straightforward application of PT to the closed shell HF wave function:

$$\psi_n^{(0)} = \Psi_0 = \Psi_{HF}.$$

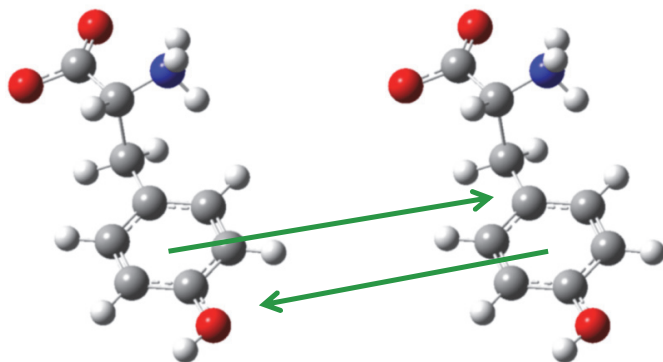
$$E \approx E(MP0) + E(MP1) + E(MP2) = E_0 + \sum_{i < j}^{occ} \sum_{a < b}^{virt} \frac{(\langle \varphi_i \varphi_j | \varphi_a \varphi_b \rangle - \langle \varphi_i \varphi_j | \varphi_b \varphi_a \rangle)^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$$

$\varepsilon_i, \varepsilon_j$ –energies of the occupied orbitals in HF wave function

$\varepsilon_a, \varepsilon_b$ –energies of the empty orbitals in HF wave function

Size Consistency and Basis Set Superposition Errors (BSSE)

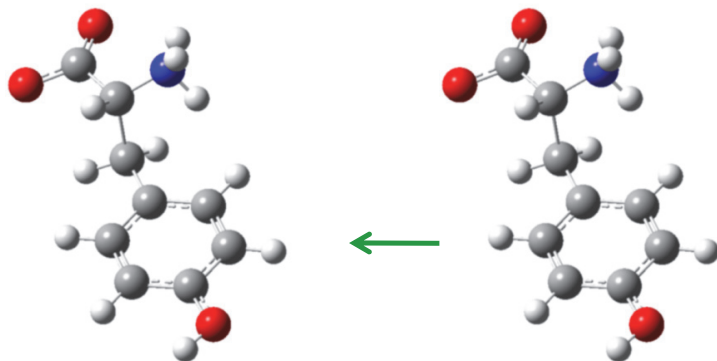
Size consistency problem in CI:



When molecules are simulated together, new excitations become available to each of them, resulting in a greater variational flexibility in the CI subspace.

$$E_{CI}(1 + 2) > E_{CI}(1) + E_{CI}(2)$$

Basis set superposition error:



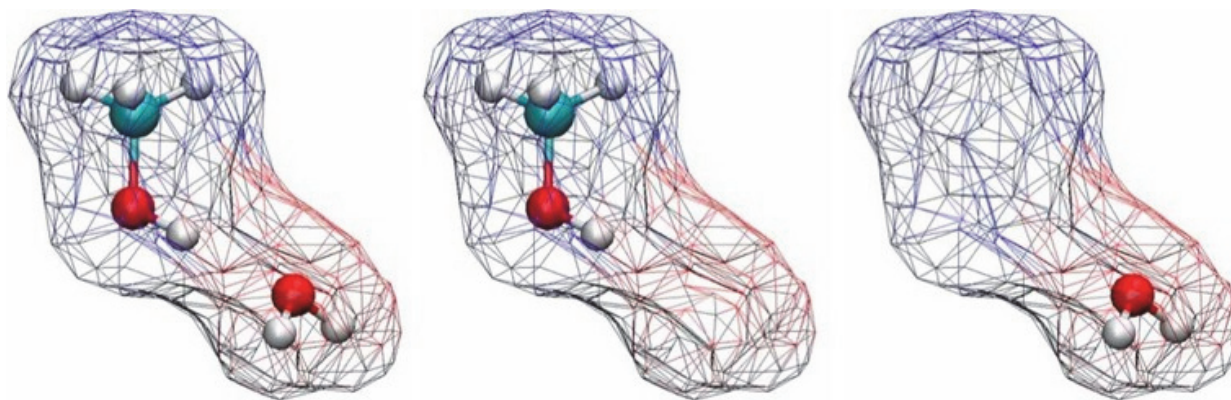
When molecules are brought close together, they augment each other's basis, thereby increasing the variational flexibility. The energy will go down even in the absence of any interaction. This results in non-physical attraction.

$$E(\textit{proximate}) < E(\textit{distant})$$

Counterpoise Correction to BSSE

Atom-centred basis sets are rarely big enough to avoid the basis set superposition error. Counterpoise correction aims to compensate this error.

1. *The energy of the complex is calculated in the usual way.*
2. *Two additional calculations are carried out for each fragment in the presence of orbitals (but not electrons or nuclei) of the other fragment.*



$$\Delta E_{\text{CP}} = E(\text{A})_{\text{AB}} + E(\text{B})_{\text{AB}} - E(\text{A})_{\text{A}} - E(\text{B})_{\text{B}}$$

Basis Set	H_0 (no CP correction) (in hartree)	Counterpoise corrected energy (in hartree)	Counterpoise BSSE Energy (in kcal/mol)
3-21G	-151.1894	-151.13462	4.066
6-31G	-151.9342	-151.93274	0.916
6-31+G(d)	151.99457	-151.99364	0.584

Summary of explicit correlation methods

N.B. Single reference methods are very accurate *if and only if* the HF reference state is a good starting point.

In systems with homolytic bond cleavage or degeneracies, MP_n , CI and CC methods can fail dramatically.

Method	$E_{\text{corr}}(\text{Be})$	Method	$E_{\text{corr}}(\text{Be})$	Method	$E_{\text{corr}}(\text{Be})$
MP2	67.85%				
MP3	86.90%	CISD	96.05%	CCSD	99.75%
MP4	94.58%			CCSD(T)	99.99%
MP5	98.15%	CISDT	96.29%	CCSDT	99.99%
MP6	99.64%				
MP7	100.15%	CISDTQ	100%	CCSDTQ	100%

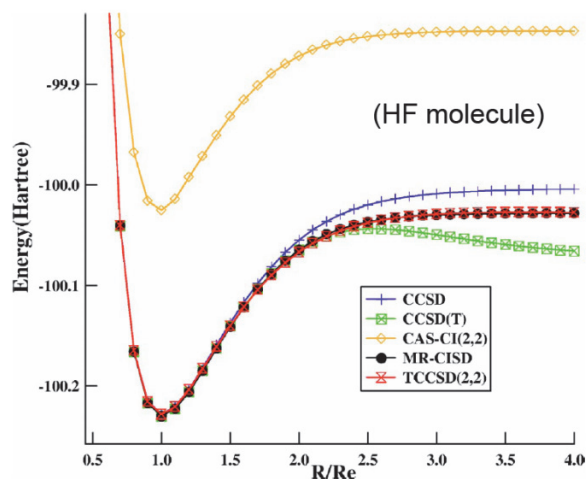


TABLE 1: RHF, B3LYP, MP2, and CISD Fully Optimized Geometries and the Experimental Values of the Product $\text{CpCo}(\text{N}(\text{O})\text{CH}_3)\text{PH}_3$ (4)

	RHF	DFT	MP2	CISD	expt
Co-Cp	2.076	1.849	1.731	1.888	1.718
Co-P	2.511	2.287	2.164	2.360	2.174
Co-N	1.991	1.772	1.711	1.787	1.780
N-O	1.226	1.324	1.378	1.295	1.282
N-C	1.487	1.514	1.527	1.504	1.484
P-Co-N	96.8	95.8	94.4	96.0	92.4
Co-N-O	124.2	126.4	128.3	125.4	125.6
Co-N-C	121.2	123.5	124.5	123.5	124.4
ave % dev	7.65	2.95	2.48	3.25	

Scaling of explicit correlation methods

Methods (without basis screening)	Scaling with the number of basis functions
HF	$O(n^4)$
MP2	$O(n^5)$
MP3, CISD, CCSD, QCISD	$O(n^6)$
MP4, CCSD(T), QCISD(T)	$O(n^7)$
MP5, CISDT, CCSDT	$O(n^8)$
MP6	$O(n^9)$
MP7, CISDTQ, CCSDTQ	$O(n^{10})$
FCI	$O(n!)$

Hartree-Fock:

- Up to a few hundred atoms.
- First and second analytical derivatives readily available with respect to common perturbations.

MP2, MP4:

- Efficient implementations scale to ~100 atoms for MP2 and to ~20 atoms for MP4.
- First and second analytical derivatives readily available with respect to common perturbations for MP2, numerical derivatives only for MP4.

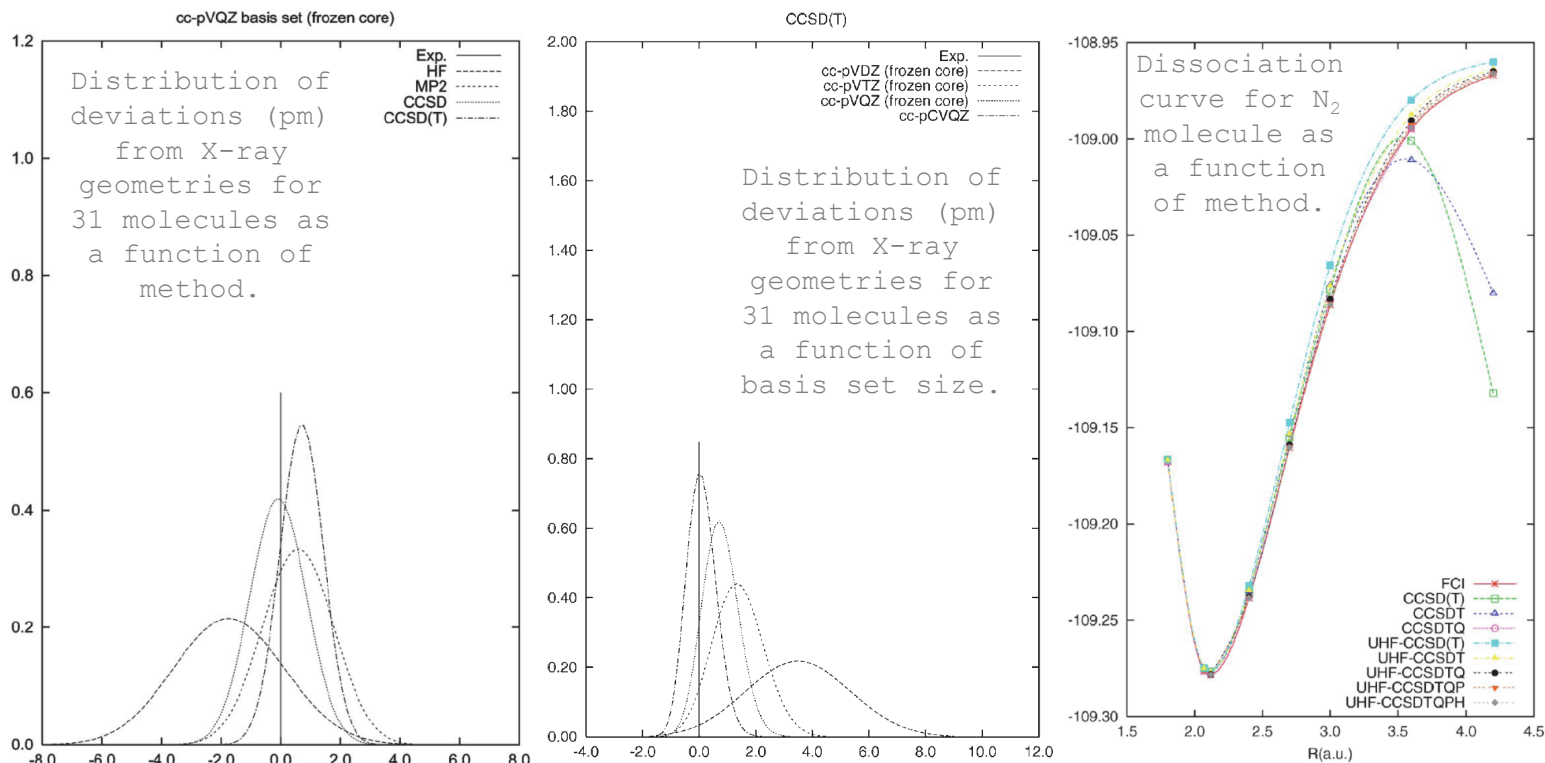
CISD, CCSD:

- Efficient implementations scale to ~20 atoms.
- First analytical derivatives usually available, second derivatives mostly numerical for now.

FCI:

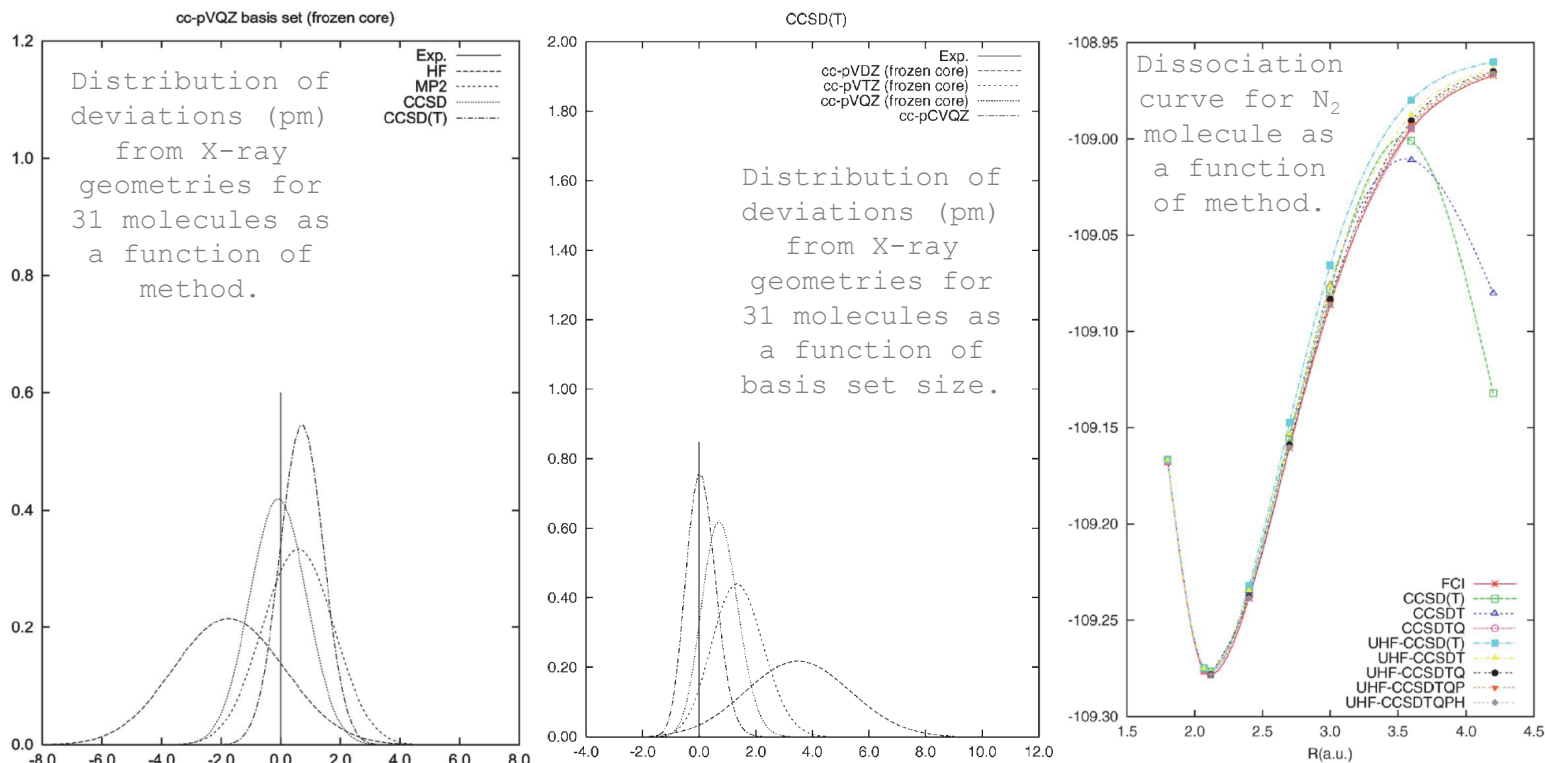
- 3-4 simple atoms at most.
- All derivatives numerical.

Performance of explicit correlation methods



For properties other than energy, a higher correlation level method does not necessarily mean a more accurate answer – beware of error compensation situations.

Performance of explicit correlation methods



For properties other than energy, a higher correlation level method does not necessarily mean a more accurate answer – beware of error compensation situations.

Correlation methods & extrapolation to the complete basis set (CBS) limit

Fast approximate methods (but capturing the essentials):

HF/6-31G(d,p) or HF/cc-pVDZ

Slow accurate methods:

MP2/6-311G(2d,2p) or MP2/cc-pVTZ

Very slow, very accurate methods:

CCSD(T)/cc-pVQZ and higher

Specialized high-accuracy compound methods:

MP2-R12, G1-G3, G2MP2, G3MP2 etc.

Energies (but not usually other parameters) can be extrapolated to the *basis set limit* – effectively to the complete basis set. Dunning-Feller extrapolation scheme is given below, in which n is the zeta index of an (aug-)cc-pV n Z basis set:

$$E_n = E_{CBS} + \alpha e^{-\beta n}$$

α, β are parameters that can be fitted for a specific case / class of molecules

Densities and distributions

Density refers to a continuous spatial distribution of a certain property.
Commonly encountered densities of interest are:

Probability density

$$p(x, y, z) = |\psi(x, y, z)|^2$$

Electron density

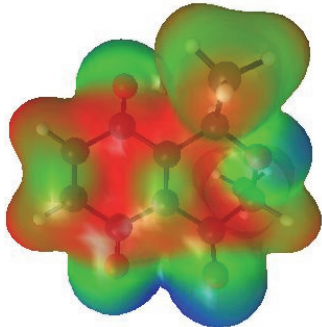
$$\rho(\vec{r}) = N \int d^3\vec{r}_2 \int d^3\vec{r}_3 \cdots \int d^3\vec{r}_N |\Psi(\vec{r}, \vec{r}_2, \dots, \vec{r}_N)|^2$$

Spin density

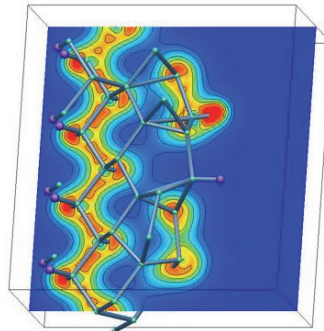
$$\sigma(\vec{r}) = \rho_{\uparrow}(\vec{r}) - \rho_{\downarrow}(\vec{r})$$

Charge density

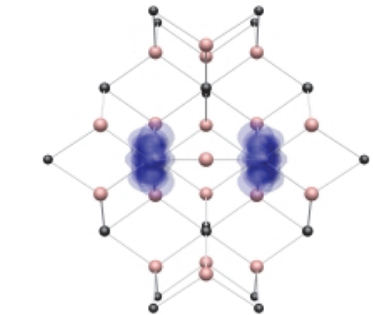
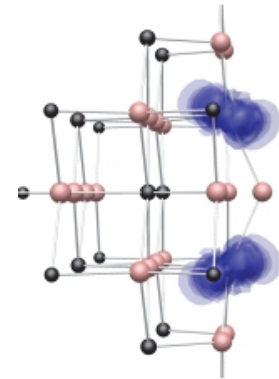
$$\rho_C(\vec{r}) = Ne \int d^3\vec{r}_2 \int d^3\vec{r}_3 \cdots \int d^3\vec{r}_N |\Psi(\vec{r}, \vec{r}_2, \dots, \vec{r}_N)|^2$$



Electron isodensity surface mapped with electrostatic potential.



Charge density map of a modified silicon surface.



Spin density around an oxygen vacancy in cerium oxide.

Practical Exercise

- 1) Open GAUSSVIEW either on your laptop or remotely on HPC
- 2) Draw a small-sized molecule, e.g. **ethane** or **ethene**
- 3) Setup a Gaussian calculation using the following computational options:
 - optimise molecular structure using **tight** convergence
 - use **spin-unrestricted DFT**, using PBE density functional
 - use **ultrafine** grids for integrals and DFT density integration
 - compute vibrational frequencies
 - check if the calculation is OK and the geometry is in the global minimum.
- 4) Open the checkpoint file with Gaussian and visualise the following quantities:
 - total electronic density,
 - spin alpha density
 - Inspect the atomic charges, bond orders, vibrational frequencies
- 5) Calculate the same molecule (just the single point energy calculation) with several basis sets and plot “total Energy = f(basis set size)”
- 6) Calculate the same molecule (just the single point energy calculation) using cc-pVTZ basis sets and the following correlation methods: MP2, MP3, MP4, CCSD, CCSD(T), CCSDT, CISD, CISDT and plot “total Energy = f(computational method)”. Use the same amount of cores for all these calculations and note the time required for each of them. Plot “total time = f(computational method)”.
- 7) Check the change in atomic charges, bond orders, etc. after the electron correlation method was considered and compare with UHF calculation done first.
- 8) Evaluate the **counterpoise correction** required for a system containing ethane+water. The oxygen of the water is to be placed at 2.0 Angstrom on any of the the four C-H-O lines, i.e. H of ethane is 2.0 Angstrom away from O of the water.